0580_w24_ms_32_Q4

Question	Answer	Marks	Partial Marks	Guidance
4(a)	2870 or 2867 to 2867.4	**2**	**M1** for $\pi \times 7.8^2 \times 15$ oe	Allow 912.6π for **2** marks Mark at Most Accurate M1 implied by 2865 to 2868.2
(b)	1350	**3**	**M2** for $\sqrt[3]{3375}^2$ oe or better or **M1** for $\sqrt[3]{3375}$ oe	Implied by 15^2 or 225 Implied by 15
(c)	$5.4 \times 100^2 = 54\,000$ or $37\,000 \div 100^2 = 3.7$	**M1**		Accept 54000 or 3.7 for M1 Ignore any units
	Area *B*	**A1**		

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Question	Answer	Marks	Partial Marks	Guidance
(d)	27.8 or 27.81… to 27.82	**2**	**M1** for sin[...] = $\frac{7}{15}$ or better or $90 - cos^{-1}(\frac{7}{15})$ oe	Accept fully complete alternative methods. note that BC = 13.26….. M1 for tan[…] = 7 / √(15^2 -7^2) oe M1 for cos[…] = √(15^2 -7^2) / 15 oe
(e)	345.6	**4**	**B3** for 28.8 **OR** **M2** for $31.2^2 – 12^2$ oe or **M1** for $[...]^2 + 12^2 = 31.2^2$ oe **M1dep** for *their* 28.8 × 12	Mark at Most Accurate M2 imp by 973.44 – 144 or 829.44 Accept alternative methods to find *EF* = 28.8 **M2** for sin($\cos^{-1}$[12/31.2]) × 31.2 or tan($\cos^{-1}$[12/31.2]) × 12 **M2** for cos($\sin^{-1}$[12/31.2]) × 31.2 or 12 ÷ tan($\sin^{-1}$[12/31.2]) **Dep**. on attempt at Pythagoras' or trig. Accept complete alternative method to find area using sine rule **M3** for 2 x ½ x 12 x 31.2 x sin($\cos^{-1}$[12/31.2]) or **M2** for ½ x 12 x 31.2 x sin($\cos^{-1}$[12/31.2])